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SEMESTER 2 EXAMINATIONS 2020/2021

MODULE: CA4024 - Building Complex Computational Models

PROGRAMME(S):
DS BSc in Data Science

YEAR OF STUDY: 4

EXAMINER(S):

Dr. Marija Bezbradica	(Internal)	(Ext:5260)
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Prof. Mathieu d'Aquin	(External)	External

TIME ALLOWED: 3 Hours

INSTRUCTIONS: **Answer 4 questions. All questions carry equal marks.**

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

QUESTION 1 ***[TOTAL MARKS: 25]***

Q 1(a) **[4 Marks]**

Deep in the redwood forest of California, dusky-footed wood rats provide up to 80% of the diet for the spotted owl, the main predator of the wood rat.

- In the absence of wood rats, the owl population would decay by 50% each year.
- In the presence of rats, the owl population will increase each year. The owls convert rat kills into population growth with an efficiency of 40%.
- In the absence of owls, the population of the rats will grow by 10% per year.
- The owls hunt the rats, so if owls are present the rat population will be reduced by 0.1 for every owl.

If $R_n, R_{n+1,}$ and O_n, O_{n+1} denote the rat and owl population in years *n* and *n+1* and

$$R_{n+1} = aR_n + bO_n$$
$$O_{n+1} = cR_n + dO_n$$

Determine the values of *a, b, c, d.*

Q 1(b) **[11 Marks]**

By setting up a matrix system for the above system of equations, find the maximum eigenvalue and associated eigenvector. Determine the long-term survival of the rat population.

Q 1(c) **[10 Marks]**

With 50% more rats killed per owl how will the above scenario in Q1(b) change? What is the critical rate of predation?

[End of Question 1]

QUESTION 2 ***[TOTAL MARKS: 25]***

Q 2(a) **[4 Marks]**

What are the two models for time series decomposition into which a series may be decomposed? For each, you should define them as a formula and say when they should be used.

Q 2(b) **[6 Marks]**

For a time series to be decomposed into one of the two models above, define clearly, in your own words, what is meant by the following, saying how they might be performed on a simple time series without cyclical component: *Detrending*, *Seasonal Index*, *Seasonally Adjusted*.

Q 2(c) **[15 Marks]**

The time series data in Table Q2 shows Quarterly Sales Data (in thousands) from 1979-81. Generate a graph of the data explaining what kind of model should be used to decompose the data and provide reasons for your choice. Calculate the Trend (using 2X4pt Moving Averages), Seasonal Index and Seasonally-Adjusted data. Graph the Trend and Seasonally-Adjusted data (but **not** the Seasonal Index data).

Year	Quarter	Sales
1979	1	72
1979	2	110
1979	3	117
1979	4	172
1980	1	76
1980	2	112
1980	3	130
1980	4	194
1981	1	78
1981	2	119
1981	3	128
1981	4	201
1982	1	81
1982	2	134
1982	3	141
1982	4	216

Table Q2

[End of Question 2]

QUESTION 3 ***[TOTAL MARKS: 25]***

Q 3(a) **[6 Marks]**

For a non-stationary time series, define clearly, in your own words, what is meant by rendering a time series (a) trend stationary and (b) difference stationary.

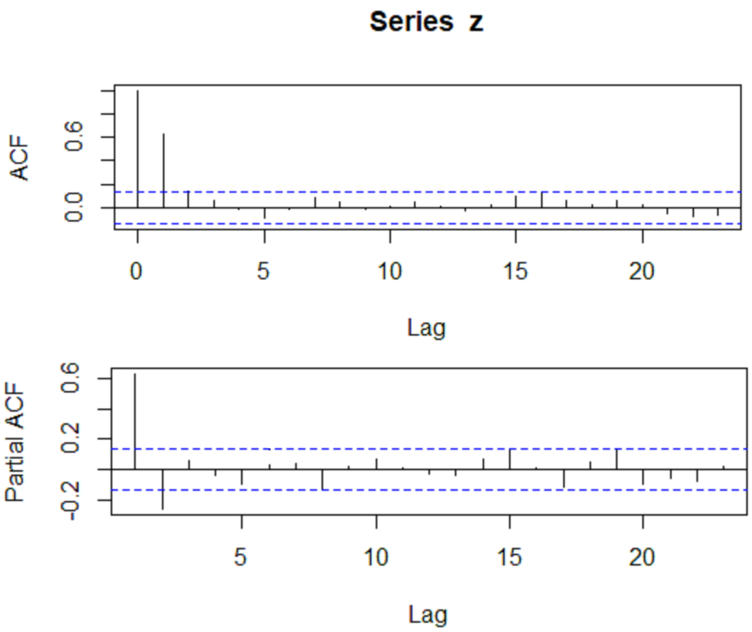
Q 3(b) **[11 Marks]**

Explain clearly and concisely what you understand by the terms *Auto-Correlation Function (ACF)* and *Partial Auto-Correlation Function (PACF)*. How may they be used to determine the values of *p*, *q* in an Autoregressive Moving Average (ARMA(*p*,*q*)) model of a time series? Show how Information Criteria may give a better way to determine these values.

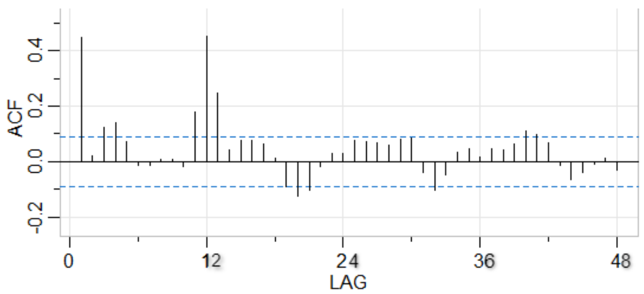
Q 3(c) **[8 Marks]**

Figure 3(c) shows 2 examples of (theoretical) Autocorrelation Function diagrams for a some ARIMA and SARIMA models. In each case, give the simplest possible form you can of the underlying model. You should justify your choice in all cases.

Case (i) Series z: ACF and PACF [4 Marks]



Case (ii) Series z: ACF and PACF [4 Marks]



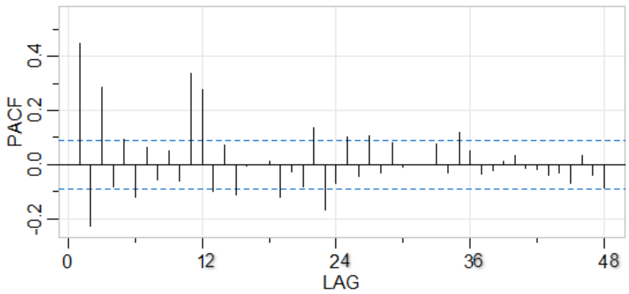


Figure Q3(c)

[End of Question 3]

QUESTION 4 ***[TOTAL MARKS: 25]***

Q 4(a) **[3 Marks]**

Define carefully, in your own words, what is meant by a *Complex System.*

Q 4(b) **[10 Marks]**

Distinguish, giving examples, between the terms *emergent properties* and *self-organization.* Is it possible for self-organization to exist without emergence?

Q 4(c) **[12 Marks]**

Describe, using an example from your experience, the steps you would take when modelling a complex system.

[End of Question 4]

QUESTION 5 ***[TOTAL MARKS: 25]***

Q 5(a) **[10 Marks]**

Compare and contrast Cellular Automata and Agent-Based models in terms of (a) What they are used for and (b) How they are constructed.

Q 5(b) **[6 Marks]**

Describe carefully, in your own words, the elements of a Cellular Automata. Your answer should show clearly labelled diagram(s) of Moore's neighbourbood and Von Neumann's Neighbourhood.

Q 5(c) **[9 Marks]**

Using Conway's Game of Life (or another such application from your experience), show how using simple rules, emergent properties can be seen in Cellular Automata models.

[End of Question 5]

QUESTION 6 ***[TOTAL MARKS: 25]***

Q 6(a) **[9 Marks]**

Distinguish clearly, using diagrams, between *Attractors, Strange Attractors* and *Limit Cycles*.

Q 6(b) **[8 Marks]**

A portion of the Logistic Map is shown in Figure Q6. Please give clear explanations for what is happening at A, B, C, D.

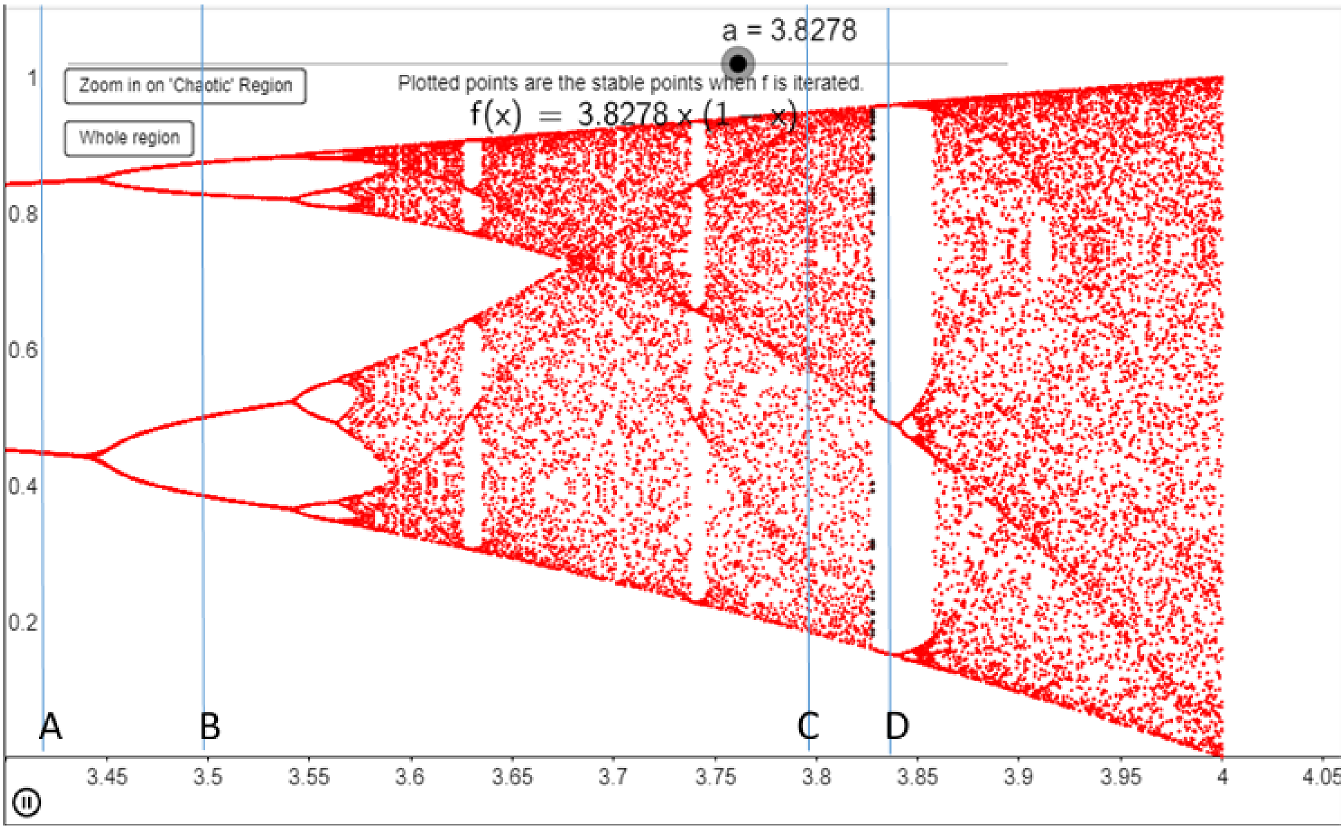


Figure Q6

Q 6(c) **[8 Marks]**

Describe, using a clearly labelled diagram, the limit cycle of the Lotka-Volterra map corresponding to the Predator-Prey system. Your answer should include a description of how one species interacts with the other.

[End of Question 6]
[END OF EXAM]